

data processing, this is an important aspect. BlockCloud provides facilities such as service-centric networking (SCN), which guarantees security and reliability of data communication.

A service-centric networking provides a mechanism for deploying replicated services, such as instances in a cloud platform with a highly distributed architecture. Here, the client requests are routed to the nearest instance to provide efficient network and infrastructure usage and improved Quality of Service (QoS). This is very useful in an IoT service based solution due to guaranteed service delivery and reliability in communication.

On the other hand, a blockchain offers decentralised network communication in order to provide reliable network transactions with the highest standard of security, where tampering of records is impossible.

Combining these two is the integrated solution offered by the BlockCloud platform, which helps a decentralised network with better network communication using SCN. Blockchain provides the highest level of security in data transmission and management in a decentralised fashion, as data transaction is controlled by multi-node communication and a shared ledger copy using smart contract management in the decentralised network service nodes.

BlockCloud with SCN for IoT solution has four major layers of implementation, as explained below.

Private edge node: This is a set of node instances that contains the instances to be used for edge caching for the encompassed services. When a service request comes to the service connector, it picks the nearest edge node for servicing the request in order to reduce the latency in network transmission.

IoT edge node: This is a group of node instances, which caches the IoT service feed in order to collect the IoT data transmission across networks. This

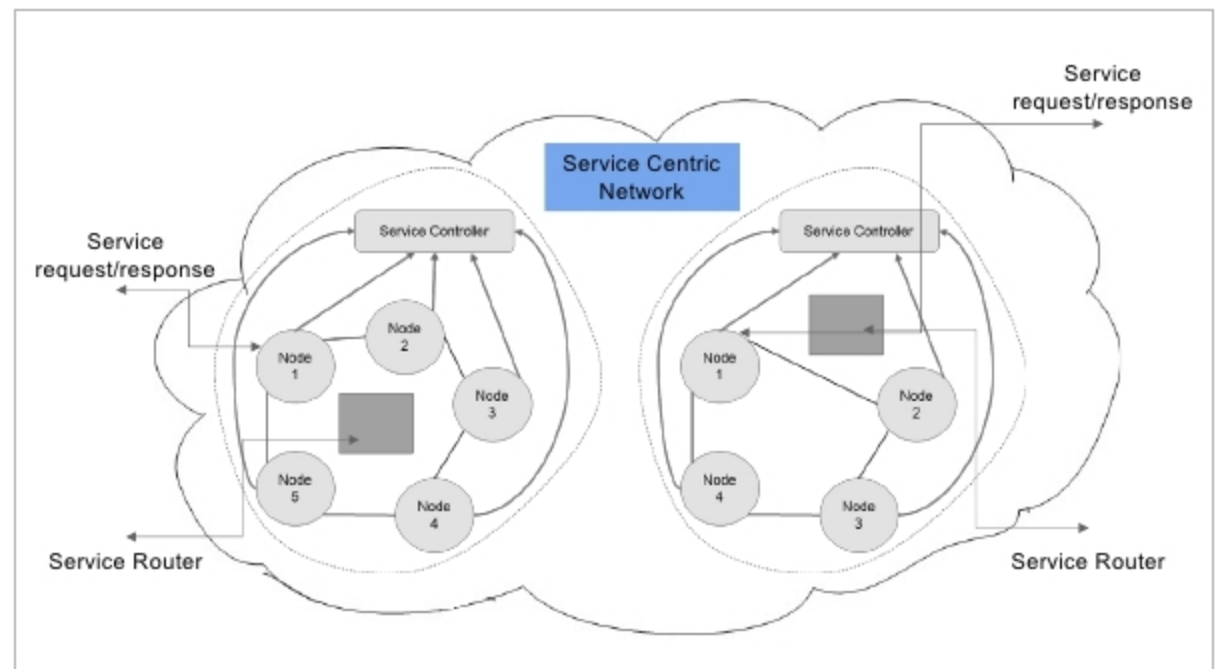


Figure 2: Service-centric networking

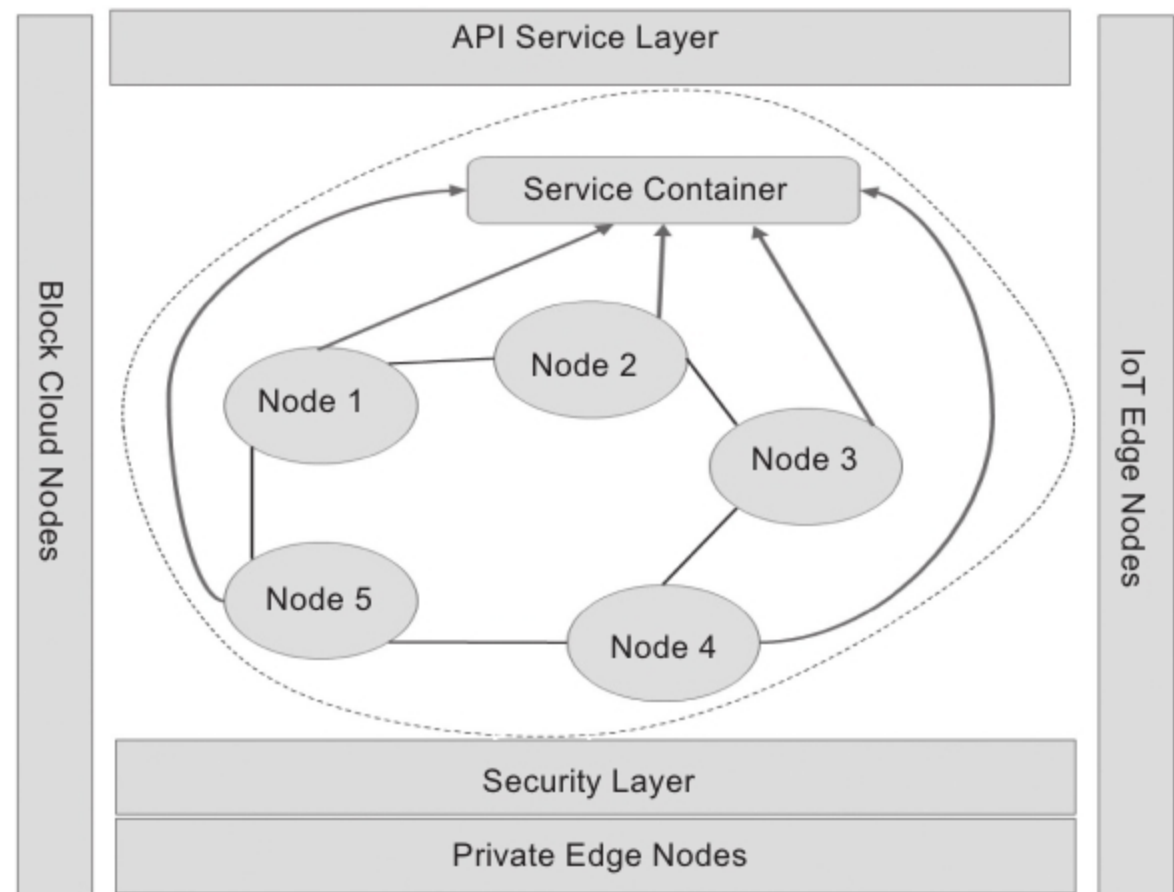


Figure 3: BlockCloud with SCN for IoT solution

helps to aggregate the data feeds in the IoT connector to be used for processing the request, when it comes from the service connector.

Security layer: This layer has the smart contracts defined for secured data transmission and data processing including application security, data security and interfacing security policies. This helps to ensure the complete network architecture is secured and that role based access control is implemented.

API service layer: This is the layer that exposes API services to the external world in order to access the service interface for request handling, monitoring requests, health-check requests, to name a few.

Some common use cases of BlockCloud

BlockCloud is commonly used in architecture where distributed network communication in a decentralised fashion can be combined in a service-